

SARASOTA BRADENTON INTL, FL, USA

WMO: 722115

Lat: 27.401N

Lon: 82.559W

Elev: 9

StdP: 101.22

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12871

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.5	(c) 6.9	(d) -3.4	(e) 2.8	(f) 10.0	(g) -0.9	(h) 3.5	(i) 10.6	(j) 11.1	(k) 20.4	(l) 10.1	(m) 19.3	(n) 2.5	(o) 40	(p) 0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 7.8	(c) 33.5	(d) 25.6	(e) 32.8	(f) 25.6	(g) 32.4	(h) 25.5	(i) 28.0	(j) 31.2	(k) 27.3	(l) 30.6	(m) 26.8	(n) 30.4	(o) 4.3	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.3	23.1	30.2	26.3	21.8	29.2	25.9	21.2	29.1	89.5	31.3	86.1	30.4	83.3	30.2	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.3	(b) 8.2	(c) 7.3	(e) 0.6	(f) 35.6	(g) 2.2	(h) 1.3	(i) -0.9	(j) 36.5	(k) -2.2	(l) 37.2	(m) -3.4	(n) 38.0	(o) -5.0	(p) 38.9		
(a) 9.3	(b) 8.2	(c) 7.3	(e) -1.3	(f) 28.2	(g) 2.4	(h) 1.0	(i) -3.0	(j) 28.9	(k) -4.4	(l) 29.5	(m) -5.7	(n) 30.1	(o) -7.5	(p) 30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.1	16.4	17.9	19.7	22.3	25.2	27.4	28.1	28.2	27.5	24.9	20.9	18.4		
	DBStd	5.06	4.37	4.06	3.47	2.44	1.85	1.36	1.18	1.10	1.31	2.67	3.28	4.25		
	HDD10.0	9	6	2	0	0	0	0	0	0	0	0	0	2		
	HDD18.3	241	88	52	28	3	0	0	0	0	0	2	15	53		
	CDD10.0	4802	204	224	302	371	473	521	563	566	526	463	327	263		
	CDD18.3	1992	28	41	71	124	214	271	304	307	276	207	92	56		
	CDH23.3	18404	110	183	382	871	1988	2839	3403	3403	2721	1711	536	256		
(13)	CDH26.7	6644	7	19	46	169	648	1156	1451	1447	1055	533	89	24		
(14)	Wind	WSAvg	3.5	3.6	3.7	4.1	4.0	3.7	3.4	3.0	3.3	3.5	3.5	3.6		
(15) Precipitation	PrecAvg	1451	75	75	91	79	68	192	220	230	216	93	53	59		
	PrecMax	1931	201	201	256	243	198	535	372	418	363	199	220	299		
	PrecMin	1099	9	19	25	27	3	50	112	114	79	25	8	12		
	PrecStd	200	48	44	64	46	52	100	62	85	80	48	43	69		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.7	28.8	29.4	31.1	33.1	34.1	34.5	34.7	34.0	32.6	30.1	28.5		
		MCWB	21.7	22.0	20.5	21.7	22.6	25.3	26.0	26.0	25.3	24.7	23.2	22.6		
	2%	DB	26.1	27.1	27.8	29.5	31.8	32.8	33.0	33.4	32.9	31.5	28.8	27.4		
		MCWB	20.8	21.6	21.0	22.0	22.9	25.4	26.0	26.1	25.2	24.4	22.8	21.9		
	5%	DB	24.5	25.8	27.0	28.5	30.9	32.2	32.5	32.7	32.3	30.7	27.7	26.2		
		MCWB	19.8	20.9	20.9	21.9	23.1	25.5	26.0	26.1	25.3	24.2	22.0	21.3		
	10%	DB	23.0	24.1	25.8	27.6	29.8	31.3	32.0	32.2	31.4	29.5	26.7	24.8		
		MCWB	19.4	19.8	20.3	21.6	23.3	25.3	25.9	26.0	25.2	23.6	21.3	20.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.4	23.8	24.4	25.2	26.8	28.5	28.9	28.8	28.7	27.3	25.4	24.4		
		MCDB	26.0	26.7	27.1	27.9	28.8	31.9	31.9	32.5	31.3	29.6	27.9	26.7		
	2%	WB	22.0	22.8	23.3	24.2	25.6	27.5	27.8	27.7	27.3	26.2	24.5	23.4		
		MCDB	24.2	25.7	26.0	27.5	28.8	30.7	31.0	31.4	30.0	29.0	27.3	25.8		
	5%	WB	21.1	21.8	22.5	23.5	25.0	26.6	27.0	27.0	26.5	25.5	23.6	22.6		
		MCDB	23.4	24.4	25.0	26.9	28.8	30.4	30.6	30.9	30.1	28.9	26.3	24.9		
	10%	WB	20.1	20.9	21.5	22.8	24.4	25.8	26.4	26.5	25.8	24.8	22.6	21.6		
		MCDB	22.5	23.5	24.1	26.2	28.2	29.9	30.4	30.6	29.7	28.3	25.1	23.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	10.0	10.0	9.7	9.2	7.9	7.6	7.8	7.8	8.8	9.8	9.7		
		MCDBR	10.4	10.2	10.6	10.5	10.5	8.9	8.4	8.6	8.6	8.9	9.8	9.5		
		MCWBR	5.8	5.5	5.1	4.7	4.1	3.6	3.4	3.4	3.3	3.5	4.8	5.1		
	5% WB	MCDBR	8.8	8.6	8.0	8.5	8.0	7.5	7.6	7.8	7.6	7.5	8.0	8.1		
		MCWBR	6.0	5.4	5.0	4.9	4.0	3.7	3.7	3.7	3.6	3.6	4.8	5.4		
(40) Clear-Sky Solar Irradiance	taub		0.344	0.358	0.369	0.393	0.422	0.445	0.467	0.447	0.422	0.391	0.355	0.347		
	taud		2.530	2.500	2.454	2.355	2.331	2.348	2.288	2.361	2.395	2.465	2.509	2.548		
	Ebn at Noon		904	917	923	905	873	848	828	845	860	870	883	885		
	Edh at Noon		90	100	111	126	129	126	134	124	116	103	92	86		
(44) All-Sky Solar Radiation	RadAvg		3.62	4.39	5.51	6.58	6.97	6.32	6.03	5.77	5.23	4.83	3.95	3.35		
	RadStd		0.27	0.28	0.43	0.29	0.37	0.52	0.36	0.35	0.34	0.29	0.20	0.27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	-4	-72	+281	+213
(47)	Regional (30 neighbors)	+0.43	+0.86	N/A	N/A	N/A	N/A	-3	-43	+169	+117

Nomenclature: See separate page